

Agent-Based Simulation and System Dynamics

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- In this chapter we discuss types of simulation other than traditional Discrete Event Simulation (DES).
- The emphasis will be on dynamic simulation models, which represent a system as it evolves over time. There are two major types of dynamic simulation models, namely, DES and continuous simulation.
- Recall that in a DES the state variables change instantaneously at a countable number of separate points in time. These points in time are the ones at which an event occurs, where an event is (usually) an occurrence that immediately changes the state of the system (e.g., an arrival or a departure).

AGENT-BASED SIMULATION

- An agent is an autonomous “entity” that can sense its environment, including other agents, and use this information in making decisions. Agents have attributes and a set of basic if/then rules that determine their behaviors.
- They may also learn (gain a better understanding of the status of other agents and their environment) and adapt their behaviors (change their decision rules) over time, which will require them to have some form of memory.
- We define an agent-based simulation to be a DES where the entities (agents) do, in fact, interact with other entities and their environment in a major way. We say that an ABS is a variation of DES, since in virtually all existing ABS, state changes do occur at a countable number of points in time.

- Sometimes, ABS takes a bottom-up approach to modeling, where the emphasis is on describing the behavior and interaction of the individual agents. Complex ABS are usually implemented in object-oriented software (e.g., Java), where instance variables (or data) correspond to attributes and methods correspond to behaviors.
- The interactions of the “low-level” agents over time result in emergent behavior of the system as a whole, which is not deducible from the characteristics of the individual agents.